



The Younger Dryas

Exploring the various causes of the Younger Dryas about 14,500 years ago
<https://digit.in/sep21-27>



Review of the evidence

An overview of all the evidence so far for and against the impact event
<https://digit.in/sep21-28>



Did a comet break up in the skies at the dawn of civilization?

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ince the last glacial maximum about 30,000 years ago, the world had been getting steadily

warmer, but there was an abrupt reversal in the trend between 12,800 and 11,600 BP, when the glaciers stopped suddenly stopped receding and began to grow again. There was an abrupt cooling of global temperatures, with the northern hemisphere becoming cold and dry. Snow and ice sheets accumulated in the Arctic and surrounding regions. This period is known as the Younger Dryas, and after a short period of global cooling, the world suddenly became warmer, and wetter as well. The Younger Dryas is known after the cold weather

flower, *Dryas octopetala*, which became widespread in the Northern Hemisphere during this brief interlude. From Asia, to North America to the Mediterranean, the sudden changes had a big impact on how human communities lived, with evidence suggesting that this sudden cooling guided early human settlements towards agriculture, as the widespread cooling prevented wild corn from growing in sufficient quantities to feed human populations. Shortly after the Younger Dryas, agricultural human civilisation cropped up in the "Fertile Crescent", between the Persian Gulf and the Red Sea, along the banks of the Tigris, the Euphrates and the Nile. In North America, the Clovis culture disappeared, along with the distinctive spear point that they are named after. Yaks, mammoths, smilodons, giant tapirs, and giant sloths went extinct in North America, along with a number of other pleistocene megafauna. In North America, tools associated with the Clovis culture along with the bones of

the animals they hunted are found below a thick layer of carbon around that time, known as black mats, indicative of widespread fires. Above this black layer, both the bones and the tools disappear. According to a controversial theory first proposed in 2007, there were a series of cometary fragments that impacted with or exploded in the atmosphere, leading to widespread fires, affecting human populations and animals around the world. This apocalyptic climatic event is known as the Younger Dryas Impact, and it is fiercely contested in the scientific community whether or not there was such an impact at all.

According to the hypothesis, the Younger Dryas impact led to a global cooling of the climate, and contributed to the disappearance of North American megafauna. To be clear this is not a simplistic story as the meteor impact that supposedly caused the extinction of the dinosaurs, which is itself under scrutiny now, with volcanic activity that led to the creation of the Deccan Traps believed to have at least contributed to the extinction.